



INTEGRATED ROBUST SOLUTIONS, ISLAMABAD

Week 7 Task: AI Internship

Advanced Machine Learning Techniques

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Prepared for IR Solutions AI Interns

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1 Introduction

This document outlines the Week 7 tasks for AI interns at IR Solutions. The tasks focus on advancing expertise in machine learning (ML) through theoretical learning of advanced ML techniques and practical implementation. Interns will study selected modules from Google's Machine Learning Crash Course and complete two programming tasks in Python using Google Colab: a Random Forest project for loan repayment prediction and a k-Nearest Neighbors (k-NN) based recommendation system for books. Weekly deliverables include a PDF report and Colab .ipynb files, submitted by **Friday, August 22, 2025, 5:30 PM PKT**.

2 Objectives

The Week 7 tasks aim to:

- Develop skills in unsupervised learning through clustering techniques to identify patterns in data.
- Understand the principles of generative models, specifically Generative Adversarial Networks (GANs), for creating realistic data.
- Gain hands-on experience with decision forests for classification tasks using the LendingClub dataset.
- Explore recommendation systems using collaborative filtering techniques on a book dataset.
- Practice professional documentation through structured PDF reports and well-commented Colab notebooks.

3 Theoretical Learning

Interns are required to complete the following modules from Google's Machine Learning Crash Course (<https://developers.google.com/machine-learning/crash-course>):

3.1 Clustering

- **Course Material:** Clustering
- **Learning Objectives:**
 - Understand clustering as a key unsupervised machine learning strategy to group related items based on similarity.
 - Learn about common clustering algorithms, such as k-means, and their applications in pattern discovery.
 - Explore evaluation metrics for clustering, such as silhouette score, and considerations for choosing the number of clusters.

3.2 Generative Adversarial Networks

- **Course Material:** Generative Adversarial Networks
- **Learning Objectives:**
 - Learn the fundamentals of GANs, which pair a generator and a discriminator to create new data instances resembling training data.
 - Understand the adversarial training process where the generator aims to "fool" the discriminator, and the discriminator learns to distinguish real data from generated data.
 - Explore practical applications of GANs, such as generating photorealistic images, and challenges in training stable GAN models.

4 Practical Tasks

Interns will complete two programming tasks in Python using Google Colab. Each task includes a dataset and specific questions to guide implementation and analysis. Submit the Colab notebooks as `Week7_InternName_Task01.ipynb` and `Week7_InternName_Task02.ipynb`.

4.1 Task 01: Random Forest Project

- **Description:** Build and evaluate Decision Tree and Random Forest models to predict whether a borrower will fully repay a loan using the LendingClub dataset (2007–2010). The dataset (`loan_data.csv`) is available in the shared Google Drive folder.
- **Steps:**
 - Load and explore the dataset using `pandas (info(), describe(), head())`.
 - Perform exploratory data analysis (EDA) with visualizations:
 - * Histogram of FICO scores by `credit.policy`.
 - * Histogram of FICO scores by `not.fully.paid`.
 - * Countplot of loans by `purpose`, colored by `not.fully.paid`.
 - * Jointplot of FICO score vs. interest rate.
 - * `lmpplot` of interest rate vs. FICO score, separated by `not.fully.paid` and `credit.policy`.
 - Preprocess the data:
 - * Convert the categorical `purpose` column to dummy variables using `pd.get_dummies`.
 - * Split the dataset into training (70%) and test (30%) sets using `train_test_split` with `random_state=101`.
 - Train Decision Tree and Random Forest models using `scikit-learn`.
 - Evaluate models using accuracy, precision, recall, F1-score, and confusion matrix; generate visualizations.
 - Save the best model using `pickle`.
- **Questions:**
 1. Load the dataset using `pd.read_csv`. How many rows and columns are in the dataset? List the column names.

2. Identify and confirm there are no missing values. Why is this important for model training?
3. Convert the `purpose` column to dummy variables. How many new columns were added, and why was `drop_first=True` used?
4. Split the dataset into training and test sets. What are the sizes of these sets?
5. Train Decision Tree and Random Forest models. Report their accuracy, precision, recall, and F1-score. Which model performed better, and why?
6. Generate and interpret the confusion matrix for the Random Forest model. What are the implications of the recall score for the minority class (`not.fully.paid=1`)?
7. Plot feature importance for the Random Forest model. Identify the top three features and explain their significance.
8. Perform 5-fold cross-validation for the Random Forest model. Report the mean accuracy and standard deviation.
9. Discuss potential improvements to address class imbalance observed in the dataset.
10. Save the Random Forest model using `pickle`. Why is this step important for reproducibility?

- **Deliverables:**

- Colab notebook (`Week7_InternName_Task01.ipynb`) with answers to the questions in Markdown cells, complete code, and visualizations (e.g., confusion matrix, feature importance).
- Include comments following PEP 8 standards and ensure the notebook is executable.

4.2 Task 02: Recommendation System Project

- **Description:** Build a k-Nearest Neighbors (k-NN) based recommendation system to suggest books based on the Goodreads dataset, accessible via Kaggle (`books.csv`).

- **Steps:**

- Load and explore the dataset using `pandas` (`info()`, `describe()`, `head()`).
- Perform EDA with visualizations:
 - * Bar plot of top 10 books with over 1M ratings, sorted by `average_rating`.
 - * Bar plot of top 10 authors by number of books.
 - * Bar plot of top 10 most-rated books by `ratings_count`.
 - * Countplot of books by `language_code` (top 10 languages).
 - * Bar plot of top 10 highly rated authors (books with `average_rating > 4.4`).
 - * Bar plot of top 10 publishers by number of books.
 - * Distribution plot of `average_rating`.
 - * Relationship plots (`relplot`) of `average_rating` vs. `ratings_count`, `num_pages`, `language_code`, and `text_reviews_count`.
- Preprocess the data:
 - * Create a `rating_between` column categorizing `average_rating` into bins (0–1, 1–2, 2–3, 3–4, 4–5).
 - * Convert `rating_between` and `language_code` to dummy variables.

- * Combine dummy variables with `average_rating` and `ratings_count` into a feature matrix.
 - * Scale the feature matrix using `MinMaxScaler`.
 - Build a k-NN model (`n_neighbors=6, algorithm='ball_tree'`) and fit it to the scaled features.
 - Implement a `BookRecommender` function to return recommended book titles.
 - Test the recommender with two book titles and analyze the results.
 - Save the model using `pickle`.
- **Questions:**
 1. Load the dataset using `pd.read_csv`. What are the shapes of the dataset, and what are the column names?
 2. Confirm there are no missing values. Why is this critical for the recommendation system?
 3. Create the `rating_between` column. How many categories were created, and why is this feature useful?
 4. Convert `rating_between` and `language_code` to dummy variables. How many new columns were added?
 5. Apply `MinMaxScaler` to the feature matrix. Why is scaling important for k-NN?
 6. Build and fit the k-NN model. Why was `algorithm='ball_tree'` chosen?
 7. Test the `BookRecommender` function with two book titles. Are the recommendations reasonable? Explain.
 8. Visualize the distribution of distances returned by `model.kneighbors` for one book. What does this indicate about the recommendation quality?
 9. Perform a sensitivity analysis by testing the k-NN model with `n_neighbors=3` and `n_neighbors=10`. How do the recommendations differ?
 10. Save the k-NN model using `pickle`. Why is this step important for reproducibility?
 - **Deliverables:**
 - Colab notebook (`Week7_InternName_Task02.ipynb`) with answers to the questions in Markdown cells, complete code, and visualizations (e.g., distribution plots, bar plots).
 - Include comments following PEP 8 standards and ensure the notebook is executable.

5 Submission Guidelines

5.1 PDF Report

- **Format:**
 - **Title Page:** Include IR Solutions logo (if available), intern's name, designation (AI Intern), department (Artificial Intelligence), and date.
 - **Sections:**
 - * **Theoretical Understanding:** Summarize key concepts from the assigned course modules (Clustering, Generative Adversarial Networks).

- * **Project Work:** Describe the two tasks, including methodology, code implementation, and results.
 - * **Challenges and Approaches:** List challenges faced in the tasks and solutions applied.
 - * **Results and Analysis:** Present model performance (e.g., accuracy, precision, recall for Random Forest; recommendation quality for k-NN) with plots (e.g., confusion matrix, feature importance, distribution plots).
 - * **Spare Time Activities:** Note any additional learning or exploration during spare time.
 - * **Conclusion:** Summarize key takeaways and plans for improvement.
 - * **References:** Cite all resources used.
- Use professional formatting with headings, subheadings, bullet points, and tables/plots for results.
- **File Naming:** `Week7_InternName_Report.pdf`

5.2 Colab Notebooks

- Submit two notebooks: `Week7_InternName_Task01.ipynb` and `Week7_InternName_Task02.ipynb`.
- Follow PEP 8 standards, include clear comments, and use print statements for debugging.
- Use Markdown cells for task overview, methodology, question answers, and analysis.
- Include visualizations (e.g., confusion matrices, feature importance, distribution plots, bar plots).
- Ensure notebooks are fully executable with no errors.

5.3 Submission Process

- Upload the PDF report and both .ipynb files to the designated Google Drive folder by **Friday, August 22, 2025, 5:30 PM PKT**.
- Ensure files are clearly named and organized in the Week 7 folder.

6 Evaluation Criteria

Interns will be evaluated based on:

- **Theoretical Understanding (30%):** Depth and clarity of concepts learned from the course modules.
- **Project Output & Code Implementation (30%):** Correctness, efficiency, and documentation of code; quality of analysis and visualizations.
- **Report Quality (20%):** Clarity, completeness, and adherence to the defined format.
- **Best Practices (10%):** Use of PEP 8, modular code, and clear documentation.
- **Participation (10%):** Engagement in the Week 7 AI standup meeting and responsiveness to feedback.

7 Additional Notes

- **Dataset Access:** The LendingClub dataset (`loan_data.csv`) is available in the shared Google Drive folder. The Goodreads dataset (`books.csv`) is accessible via Kaggle.
- **Support:** Reach out to AI team members or Lead AI via Slack or during the standup meeting (Friday, 3:00 PM PKT) for guidance.
- **Expectations:** Maintain professionalism, meet deadlines, and actively engage in learning and collaboration.

8 References

- Google Machine Learning Courses: <https://developers.google.com/machine-learning>
- GeeksforGeeks:
 - <https://www.geeksforgeeks.org/machine-learning>
 - <https://www.geeksforgeeks.org/machine-learning-projects>
 - <https://www.geeksforgeeks.org/deep-learning-tutorial>
 - <https://www.geeksforgeeks.org/computer-vision>
 - <https://www.geeksforgeeks.org/natural-language-processing-nlp-tutorial>
- YouTube Tutorials:
 - https://www.youtube.com/watch?v=_3y7C7_do-k
 - <https://www.youtube.com/watch?v=7Bg3iBq-SIY>
 - <https://www.youtube.com/watch?v=0YdpwSYMY6I>
 - <https://www.youtube.com/watch?v=yLlgHKaMyIw>
 - <https://www.youtube.com/watch?v=NOJOYcmyDhM>